RACHEL N. SLAYBAUGH

DCVC ⋄ 270 University Ave ⋄ Palo Alto, CA 94301
rachel.slaybaugh@dcvc.com ⋄ (415) · 841 · 3935

EXPERIENCE & ACCOMPLISHMENTS

Data Collective Venture Capital (DCVC) Feb. 2022 – present
Partner *Palo Alto, CA*

- Source, select, fund, and support deep tech companies in the climate space
- Board member for Radiant (independent, 04/22-11/24; investor, 11/24-present), Fervo Energy (04/24-present), Fourth Power (11/23-present), and Equilibrium Energy (12/23-present); Board observer for Brimstone (03/22-present), Fervo (08/22-04/22), and Verdigris (08/23-present)

Lawrence Berkeley National Laboratory Jan. 2021 – Dec. 2021
Cyclotron Road Division Director *Berkeley, CA*

- Sourced and selected ∼10 hard tech innovators per year for a 2-year fellowship program to turn their technology concept into a product with a positive societal impact (∼$6M/year)
- Fellows embed at LBNL where we support their technical development and collaborations
- Responsible for outcomes, safety, and reporting to Department of Energy and LBNL leadership

Advanced Research Projects Agency – Energy Jan. 2017 – Oct. 2020
Program Director *Washington, DC*

- Created programs and selected teams developing technologies for advanced nuclear fission reactors, $95M across 28 teams: MEITNER, the Nuclear OPEN+ cohort, LISE, and GEMINA
- Provided deep technical and business guidance while managing: the nuclear teams; the TERRA and ROOTS Programs, supporting research for sensing and data analytics for above- and below-ground plant outcomes (∼$90M across 18 teams); and the FOCUS Program, supporting research for solar technologies that combine photovoltaic and concentrated solar power technologies (∼$12M across 4 teams)
- Supported program creation in other energy areas through workshops, brainstorming, and feedback

University of California, Berkeley Jan. 2014 – Dec. 2021
Associate Professor of Nuclear Engineering *Berkeley, CA*

- Founded the Nuclear Innovation Bootcamp, which brings diverse students from around the world to learn skills essential to innovation in nuclear energy
- Developed numerical methods for neutral particle transport with an emphasis on supercomputing and advanced architectures; applications in reactor design, shielding, and nonproliferation
- Also mentored PhDs in optimization, thermal fluids, and cryptography and anomaly detection
- Published 25 journal articles, 44 refereed conference proceedings, 3 technical reports, 2 book chapters, 5 open source pieces of software, and 2 policy pieces
- Graduated 9 PhD and 4 MS students; research adviser for 1 assistant project scientist, 1 postdoctoral scholar, 1 visiting scholar, and 15 undergraduate students
- Won >$2.5M as principal investigator (PI) and >$26M as co-PI for numerical methods research
- Created and supported a course in which Berkeley students guide second graders at under-served elementary schools in Oakland in hands on science experiments

Bettis Laboratory Mar. 2012 – Aug. 2014
Senior Engineer in the Shield Design and Development group *West Mifflin, PA*

- Implemented and developed several methods for variance reduction in Monte Carlo neutral particle transport that made problems perviously too hard to solve with this high-accuracy tool solvable
- Qualified these methods and software for use in shield design to dramatically reduce time and improve accuracy in design calculations

University of Wisconsin–Madison Sept. 2006 – Nov. 2011
Research Assistant / Rickover Fellow *Madison, WI*

- Dissertation: "Acceleration Methods for Massively Parallel Deterministic Transport" where I added 3 new methods to Denovo, software from Oak Ridge National Laboratory, that are still used
- Developed two Monte Carlo source sampling methods for arbitrarily shaped plasma sources

Penn State Breazeale Reactor Aug. 2003 – Apr. 2006
Reactor Operator *University Park, PA*

- NRC licensed Reactor Operator for TRIGA Mark III reactor; calibrated gamma irradiator

EDUCATION

Ph.D.	**University of Wisconsin–Madison**, Nuclear Engineering and Engineering Physics with a certificate in Energy Analysis and Policy	2011
M.S.	**University of Wisconsin–Madison**, Nuclear Engineering and Engineering Physics	2008
B.S.	**Pennsylvania State University**, Nuclear Engineering	2006

LEADERSHIP & SERVICE

Non-Profit Boards and Leadership

SCSP Commission on the Scaling of Fusion Energy	2024-present
LabStart Advisor	2022-present
National Renewable Energy Laboratory Investor Advisory Board (Chair 2024)	2022-present
National Academies of Science member of the Committee on Laying the Foundations for New and Advanced Nuclear Reactors in the United States	2020-2022
Activate Leadership Council	2021-present
Biden-Harris Transition Team, Department of Energy Agency Review Team	2020
Good Energy Collective, Founding Board Chair and Board Member	2020-present
Nuclear Science and Engineering Editorial Advisory Board	2020-2023
University of Michigan, NERS Department Advisory Board	2019-2021
Nuclear Energy Advisory Committee (a U.S. FACA), Appointed Member	2016-2017

Software and Computing: Berkeley Institute for Data Science, Berkeley Research Computing, Software & Data Carpentry, The Hacker Within

American Nuclear Society: Held many leadership roles 2007-2019

Reviewer for US DOE ARPA-E, Fusion Energy Sciences, Technology Commercialization Fund, Canadian Innovation Fund, and a number of journals and technical conferences

AWARDS

Pennsylvania State University Outstanding Engineering Alumni Award	2024
Worth Groundbreaking Women Award	2024
University of Wisconsin–Madison College of Engineering Early Career Achievement Award	2023
American Nuclear Society (ANS) Young Member Excellence Award	2014
Rickover Fellowship	2008-2011